

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JNANA SANGAMA”, BELAGAVI - 590018



DEEP LEARNING MINI PROJECT REPORT(CS722T2C) on

“Driver Drowsiness Detection Using CNN”

Submitted by

Nandan V Poojary

4SF23CS126

Rahul K U

4SF23CS163

Rohit L

4SF23CS171

In partial fulfillment of the requirements for the VII semester

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE & ENGINEERING

Under the Guidance of

Ms. Shiji Abraham

Assistant Professor

Department of CSE

at



SAHYADRI

College of Engineering & Management

An Autonomous Institution

MANGALURU

2026- 27



EVALUATION RUBRICS

Name: Nandan V Poojary

USN: 4SF23CS126

Sl. No.	Evaluation Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)	Marks
1	Problem Definition & Objectives	Problem is clearly defined, relevant, and objectives are specific and measurable	Problem and objectives are clear with minor gaps	Problem is partially defined; objectives need improvement	Problem/objectives are unclear	
2	Technical Approach & Implementation	Appropriate methodology, algorithms/tools used effectively; implementation is complete	Appropriate technical approach with minor issues	Basic implementation with limited technical depth	Incomplete or inappropriate implementation	
3	Innovation / Application / Problem Solving	Demonstrates strong originality, creativity, and effective solution to the identified problem	Shows some innovation and good problem-solving	Limited innovation; mostly conventional approach	No significant innovation/problem-solving	
4	Results, Testing & Analysis	Results are clearly presented, thoroughly tested, and critically analyzed using suitable measures	Results presented with adequate testing and analysis	Basic results and limited testing/analysis	Results are incomplete or unsupported	
5	Documentation, Presentation & Viva	Well-structured report, clear presentation, excellent communication and confident viva	Good documentation, presentation and satisfactory viva	Adequate documentation/presentation with limited clarity	Poor documentation, presentation or viva	
	Total					

Sl. No.	Evaluation Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)	Marks
1	Problem Definition & Objectives	Problem is clearly defined, relevant, and objectives are specific and measurable	Problem and objectives are clear with minor gaps	Problem is partially defined; objectives need improvement	Problem/objectives are unclear	
2	Technical Approach & Implementation	Appropriate methodology, algorithms/tools used effectively; implementation is complete	Appropriate technical approach with minor issues	Basic implementation with limited technical depth	Incomplete or inappropriate implementation	
3	Innovation / Application / Problem Solving	Demonstrates strong originality, creativity, and effective solution to the identified problem	Shows some innovation and good problem-solving	Limited innovation; mostly conventional approach	No significant innovation/problem-solving	
4	Results, Testing & Analysis	Results are clearly presented, thoroughly tested, and critically analyzed using suitable measures	Results presented with adequate testing and analysis	Basic results and limited testing/analysis	Results are incomplete or unsupported	
5	Documentation, Presentation & Viva	Well-structured report, clear presentation, excellent communication and confident viva	Good documentation, presentation and satisfactory viva	Adequate documentation/presentation with limited clarity	Poor documentation, presentation or viva	
	Total					

Sl. No .	Evaluation Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)	Marks
1	Problem Definition & Objectives	Problem is clearly defined, relevant, and objectives are specific and measurable	Problem and objectives are clear with minor gaps	Problem is partially defined; objectives need improvement	Problem/objectives are unclear	
2	Technical Approach & Implementation	Appropriate methodology, algorithms/tools used effectively; implementation is complete	Appropriate technical approach with minor issues	Basic implementation with limited technical depth	Incomplete or inappropriate implementation	
3	Innovation / Application / Problem Solving	Demonstrates strong originality, creativity, and effective solution to the identified problem	Shows some innovation and good problem-solving	Limited innovation; mostly conventional approach	No significant innovation/problem-solving	
4	Results, Testing & Analysis	Results are clearly presented, thoroughly tested, and critically analyzed using suitable measures	Results presented with adequate testing and analysis	Basic results and limited testing/analysis	Results are incomplete or unsupported	
5	Documentation, Presentation & Viva	Well-structured report, clear presentation, excellent communication and confident viva	Good documentation, presentation and satisfactory viva	Adequate documentation/presentation with limited clarity	Poor documentation, presentation or viva	
	Total					



SAHYADRI
COLLEGE OF ENGINEERING & MANAGEMENT
An Autonomous Institution
MANGALURU

Department of Computer Science & Engineering

DECLARATION

We hereby declare that the entire work embodied in this Mini Project Report titled “**Driver Drowsiness Detection Using CNN**” has been carried out by us at Sahyadri College of Engineering & Management, Mangaluru under the supervision of **Ms. Shiji Abraham**, in partial fulfillment of the requirements for the VII semester of **Bachelor of Engineering in Computer Science & Engineering**. This report has not been submitted to this or any other University for the award of any other degree.

Nandan V Poojary (4SF23CS126)

Rahul K U (4SF23CS163)

Rohit L (4SF23CS171)

Dept. of CSE, SCEM, Mangaluru.

Abstract

Driver drowsiness is a major cause of road accidents, particularly during long-distance and nighttime driving. Continuous monitoring of driver alertness can help identify early signs of drowsiness and provide timely warnings. This project presents a real-time Driver Drowsiness Detection System using Deep Learning, Computer Vision and CNN, designed to monitor facial and eye characteristics through a webcam without physical sensors or wearable devices.

The proposed system integrates a Convolutional Neural Network (CNN) for facial drowsiness classification with a MobileNetV2-based eye state detection model to identify whether the driver's eyes are open or closed. OpenCV performs real-time video acquisition and face detection. For each frame, the facial region is analyzed by the CNN, while extracted eye regions are processed using MobileNetV2. The outputs are combined using a weighted score fusion approach, where facial drowsiness contributes 30% and eye-state detection contributes 70%. When the combined drowsiness score reaches or exceeds the 70% threshold, an alert is generated.

The system uses a FastAPI backend and WebSocket communication to transmit detection results to a web-based monitoring interface displaying the driver's state, drowsiness score, eye status, and alerts. The project demonstrates the application of deep learning and computer vision for non-contact, real-time driver safety monitoring.

Keywords: Driver Drowsiness Detection, Deep Learning, Convolutional Neural Network (CNN), MobileNetV2, Computer Vision, Eye-State Detection, Facial Analysis, Real-Time Monitoring, Weighted Score Fusion, OpenCV, FastAPI, WebSocket, Driver Safety.

Table of Contents

Content	Page No.
Abstract	i
Table of Contents	ii
1. Introduction 1.1 Overview 1.2 Scope & Motivation	1-2
2. Literature Survey (Refer at least 20 Most recent suitable papers)	3-5
3. Problem Formulation 3.1 Problem Description 3.2 Problem Statement 3.3 Objectives 3.4 Functional Requirements 3.5 Non-Functional Requirements	6-8
4. Project Design and Implementation 4.1 Proposed Project Architecture 4.2 High Level Design 4.3 Detailed Design 4.4 Datasets Descriptions 4.5 Tools and Technologies used for Implementation	9-16
5. Results and Discussion 5.1 Experimentation Environment set up 5.2 Functional Requirements Results achieved (Tabular & Graphical results with Descriptions) 5.3 Non-Functional Requirements Results achieved (Tabular & Graphical results with Descriptions) 5.4 Comparison of Results with Existing works (Cited in Literature Survey) 5.5 Project GUI Snapshots (Major Functionalities) 5.6 Societal Impact of the Project 5.7 SDG Mapped	17-25
Conclusion and Future scope	26
References	27-28

1. Introduction

1.1 Overview

Driver drowsiness is a significant road-safety concern that can reduce a driver's attention, reaction time, decision-making ability, and overall control of a vehicle. Drowsiness can occur due to prolonged driving, insufficient sleep, nighttime travel, monotonous driving conditions, or physical and mental fatigue. In many cases, the early signs of drowsiness are reflected through observable facial and eye characteristics, such as prolonged eye closure and changes in facial appearance. Therefore, continuous monitoring of these visual indicators can help identify potentially dangerous levels of driver fatigue and provide timely warnings. Traditional driver monitoring approaches may rely on physical sensors, wearable devices, physiological measurements, or manually defined visual features. Although such methods can provide useful information, they may introduce additional hardware requirements or cause inconvenience to the driver. Recent developments in deep learning and computer vision have enabled camera-based systems to automatically learn relevant features from facial images and video frames. Convolutional Neural Networks (CNNs), in particular, have demonstrated their ability to perform image classification tasks by learning spatial features directly from input images. Lightweight architectures such as MobileNetV2 can also be used for efficient classification of specific facial regions, such as the eyes.

This project proposes a real-time Driver Drowsiness Detection System using Deep Learning and Computer Vision. The system uses a webcam to continuously capture the driver's face and applies an integrated detection pipeline consisting of OpenCV-based face detection, a CNN-based facial drowsiness classifier, and a MobileNetV2-based eye state classifier. The facial region is analyzed to obtain a face-based drowsiness score, while the detected eye regions are independently analyzed to determine whether the eyes are open or closed. The outputs from these models are combined using a weighted score-fusion mechanism, with 30% contribution from the face-based prediction and 70% contribution from the eye-based prediction. A final score of 70% or above indicates a drowsy condition and generates an alert state.

The detection system is further integrated with a FastAPI backend and WebSocket communication to transmit real-time results to a web-based monitoring dashboard. The dashboard provides information such as the driver's current state, drowsiness score, eye status, alert condition, and other system information. Thus, the project combines deep learning, computer vision, real-time processing, and web technologies into a unified driver-monitoring system.

1.2 Scope & Motivation

1.2.1 Scope

The scope of this project is to develop a real-time, non-contact driver drowsiness detection system using deep learning and computer vision techniques. The system uses a webcam to continuously capture the driver's facial information and analyze visual indicators associated with drowsiness, focusing on the overall facial region and the state of the driver's eyes. The system incorporates a Convolutional Neural Network (CNN) for facial drowsiness classification and a MobileNetV2-based model for eye-state detection. OpenCV is used for video acquisition, face detection, and extraction of facial and eye regions. The predictions are combined using weighted score fusion, with the face-based prediction contributing 30% and the eye-based prediction 70% to the final drowsiness score. When the combined score reaches or exceeds 70%, the system identifies a drowsy condition and generates an alert.

The scope also includes integration with a FastAPI backend and WebSocket communication for real-time transmission of detection results. A web-based frontend displays the driver's current alertness state, drowsiness score, eye status, alert information, and system monitoring details. Thus, the project covers the complete pipeline from video capture and deep learning inference to real-time result visualization and provides a foundation for future integration with vehicle safety systems.

1.2.2 Motivation

Driver drowsiness is a critical safety concern because reduced alertness can affect concentration, reaction time, and the ability to respond appropriately to road conditions. Drowsiness may develop gradually during prolonged driving, making continuous monitoring important for identifying early signs and providing timely warnings. The motivation behind this project is to develop an automated and non-contact approach for monitoring driver alertness using a webcam. Unlike systems that require wearable devices or specialized physiological sensors, a camera-based system can analyze facial and eye characteristics without physical interaction with the driver. Deep learning techniques such as CNN-based image classification and MobileNetV2 provide an effective approach for analyzing these visual characteristics.

The project also focuses on combining multiple visual indicators rather than relying on a single prediction source. Facial drowsiness classification provides information about the overall facial appearance, while eye-state detection provides information about whether the driver's eyes are open or closed. These outputs are combined using a weighted score-fusion approach to determine the final drowsiness level. The integration of the detection system with a web-based dashboard enables real-time monitoring and visualization. Thus, the project applies deep learning and computer vision to develop a practical, non-contact driver monitoring system for detecting drowsiness and generating timely alerts.

2. Literature Survey

Driver drowsiness detection has received considerable attention in recent years due to the need for continuous monitoring of driver alertness and the increasing availability of computer vision and deep learning techniques. Existing research has explored different approaches based on facial features, eye closure, yawning, head movement, physiological signals, and combinations of multiple visual indicators. With the development of deep learning, CNN-based methods have become widely used because they can automatically learn relevant features from facial and eye images without depending entirely on manually designed features.

El-Nabi et al. [1] presented an attention-enhanced CNN-based approach for real-time driver drowsiness detection. The study focuses on facial features and deep learning for real-time monitoring, demonstrating the applicability of CNN architectures for identifying drowsiness-related visual characteristics. The use of attention mechanisms is particularly relevant because it enables the model to focus on important regions of the input image.

Wei et al. [2] proposed a prior knowledge-guided multi-information graph convolutional network for driver drowsiness detection. Their approach combines information from multiple facial regions and uses graph-based learning to model relationships between different facial features. This demonstrates that considering relationships between multiple facial regions can provide useful information for drowsiness classification.

Abd El-Nabi et al. [3] developed a real-time drowsiness and fatigue recognition system using a customized CNN for open- and closed-eye detection. The study also incorporates an eye aspect ratio-based approach for identifying eye closure. This work is closely related to eye-state analysis because prolonged eye closure is an important visual indicator of driver drowsiness.

Beles et al. [4] investigated a multi-method driver drowsiness detection system for vehicles with autonomous driving functions. Their work combines different sources of information, including eye-related measurements and physiological information, with a decision-making mechanism. The study highlights the importance of combining multiple indicators when monitoring driver alertness.

Lin et al. [5] proposed a multi-aware graph convolutional network that considers relationships between different facial-region features for driver drowsiness detection. Their work demonstrates the use of advanced deep learning architectures for extracting and combining information from multiple regions of the driver's face.

Salem and Waleed [6] investigated real-time drowsiness detection using CNNs and transfer learning. Their work demonstrates that pretrained deep learning models can be adapted for drowsiness classification

and used in practical real-time applications. This is relevant to the proposed system because transfer-learning-based architectures can reduce the computational and data requirements associated with training deep models from scratch.

Pahariya et al. [7] specifically investigated MobileNetV2 with transfer learning for driver drowsiness detection. MobileNetV2 is designed as a lightweight deep learning architecture and is therefore suitable for applications where computational efficiency is important. This work is directly relevant to the proposed use of MobileNetV2 for eye-state classification.

Florez et al. [8] proposed a CNN-based approach for driver drowsiness detection through real-time eye-state identification. The study emphasizes the importance of identifying whether the driver's eyes are open or closed and demonstrates the suitability of CNNs for extracting eye-related visual features.

Jahan et al. [9] developed a real-time driver drowsiness detector using deep learning. Their work investigates visual characteristics related to driver alertness and demonstrates the application of deep learning for real-time drowsiness monitoring. Such approaches support the use of camera-based systems for continuous driver observation.

Rozaimi et al. [10] investigated deep learning models for driver drowsiness detection using different camera positions. The study compared architectures including MobileNetV2, ResNet50, and InceptionV3, highlighting the effect of camera placement and model selection on detection performance. This is relevant to practical deployment because camera position can influence the quality of facial and eye information available to the model.

Ahmed et al. [12] proposed an intelligent driver drowsiness detection system based on multiple CNN models and facial subsampling. The approach analyzes different facial characteristics and combines deep learning information for drowsiness classification. This supports the idea that multiple facial cues can be used together rather than relying on a single feature.

Bekhouche et al. [13] investigated driver drowsiness detection in video sequences using selected deep features. Their work emphasizes the use of deep representations extracted from video data for identifying drowsiness. Similarly, Magán et al. [14] applied deep learning techniques to sequences of images for driver drowsiness detection, demonstrating the importance of analyzing visual information across consecutive frames.

Vimala et al. [15] proposed a lightweight deep learning model for detecting driver fatigue. Lightweight models are particularly useful for real-time applications because they can reduce computational requirements while maintaining the ability to process visual information efficiently. This supports the consideration of computational efficiency in practical driver monitoring systems.

Gomaa et al. [16] proposed a CNN-LSTM-based approach for driver drowsiness prediction. The combination of CNNs for spatial feature extraction and LSTM networks for temporal information demonstrates how sequential information can be incorporated into drowsiness detection. This provides a possible direction for future improvement of systems that currently analyze individual video frames.

Irsan et al. [17] presented a driver drowsiness monitoring framework combining CNN, OpenCV, and the Internet of Things. Their system demonstrates the integration of computer vision-based detection with monitoring and communication technologies. This is particularly relevant to the proposed project, which similarly combines computer vision and deep learning with a web-based real-time monitoring interface.

Phan et al. [18] proposed a driver drowsiness detection and smart alerting system using deep learning and IoT technologies. Their work demonstrates how detection results can be connected to an alerting mechanism, emphasizing that a practical drowsiness detection system should not only classify the driver's state but also provide an appropriate warning.

Fonseca and Ferreira [19] presented a systematic review of deep learning-based driver drowsiness detection models. The review highlights the growing use of deep learning for visual driver monitoring and provides an overview of different datasets, architectures, and detection strategies used in existing research. The study provides a broader perspective on the development of deep learning-based drowsiness detection systems.

Venkateswarlu and Chirra [20] proposed a CNN-ViT-based multi-feature learning approach for driver drowsiness detection. Their work combines convolutional neural network features with Vision Transformer-based representations and considers multiple driver states. This represents the development of more advanced architectures for extracting complementary visual features.

Summary of Literature Survey

The reviewed studies show that CNN-based image classification, eye-state detection, transfer learning, lightweight architectures, and multi-feature approaches are widely used in driver drowsiness detection. Several studies emphasize eye closure as an important indicator, while others combine multiple facial features or incorporate temporal information. MobileNetV2-based approaches are particularly relevant for lightweight and real-time applications, whereas CNN-LSTM, graph convolutional networks, and CNN-ViT architectures demonstrate possible directions for advanced analysis.

Based on the reviewed literature, the proposed project focuses on a practical combination of CNN-based facial drowsiness classification and MobileNetV2-based eye-state detection. Instead of relying on a single visual indicator, the system combines the outputs of both models using a weighted score-fusion mechanism. The resulting pipeline is integrated with OpenCV for video processing and a FastAPI-WebSocket-based web interface for real-time monitoring.

3. Problem Formulation

3.1 Problem Description

Driver drowsiness is a serious safety issue that can occur due to prolonged driving, fatigue, insufficient sleep, and reduced alertness. As drowsiness develops, a driver's concentration and reaction ability may decrease, increasing the risk of accidents. Since drowsiness can occur gradually, continuous monitoring is required to identify visual signs such as changes in facial appearance and prolonged eye closure.

Traditional monitoring methods may require wearable or physiological sensors, which can increase system complexity and inconvenience. Therefore, there is a need for a camera-based, non-contact system capable of monitoring driver alertness in real time. The proposed project addresses this problem by using deep learning and computer vision to analyze facial and eye information obtained from a webcam.

3.2 Problem Statement

To develop a real-time driver drowsiness detection system that uses a webcam to capture facial information and deep learning models to identify signs of driver drowsiness. The system uses a CNN model for facial drowsiness classification and a MobileNetV2-based model for eye-state detection. The predictions from both models are combined using weighted score fusion, with 30% contribution from the face-based prediction and 70% from the eye-based prediction. When the final drowsiness score reaches or exceeds 70%, the system identifies a drowsy condition and generates an alert. The detection results are transmitted through a FastAPI backend and WebSocket connection to a web-based monitoring interface for real-time visualization.

3.3 Objectives

The main objectives of the proposed project are:

1. To develop a real-time driver drowsiness detection system using deep learning and computer vision, incorporating CNN-based facial drowsiness classification and MobileNetV2-based eye-state detection.
2. To detect and classify driver drowsiness by processing facial and eye regions from live webcam frames using OpenCV and combining the model predictions through a weighted score-fusion mechanism with a 70% drowsiness threshold.
3. To provide real-time drowsiness alerts and monitoring by integrating the detection system with a FastAPI backend, WebSocket communication, and a web-based dashboard displaying the driver's alertness state, drowsiness score, and eye status.

4. To provide a practical and extensible driver monitoring solution that can serve as a foundation for future improvements and integration with vehicle safety applications.

3.4 Functional Requirements

FR1: Webcam Video Acquisition

The system captures live video frames from the driver's webcam for continuous monitoring.

FR2: Face Detection and Region Extraction

OpenCV detects the driver's face from the captured video frames and extracts the facial region for further analysis.

FR3: Facial Drowsiness Classification

The detected facial region is processed using the trained CNN model to obtain a facial drowsiness prediction.

FR4: Eye-State Detection

The relevant eye regions are extracted from the detected face and analyzed using the MobileNetV2-based model to classify the eyes as open or closed.

FR5: Drowsiness Score Fusion

The facial and eye-based predictions are combined using the defined weighted score-fusion mechanism, with 30% contribution from the face score and 70% contribution from the eye score.

FR6: Drowsiness Decision and Alert

The final drowsiness score is compared with the 70% threshold. When the threshold is reached or exceeded, the driver is identified as drowsy and the corresponding alert condition is generated.

FR7: Real-Time Result Communication

Detection results are transmitted from the backend to the frontend through WebSocket communication.

FR8: Dashboard Visualization

The web-based monitoring dashboard displays the driver's current alertness state, drowsiness score, eye status, and alert information.

3.5 Non-Functional Requirements

NFR1: Real-Time Performance

The system should process live webcam frames and provide drowsiness detection results with minimal delay to support continuous driver monitoring.

NFR2: Prediction Reliability

The trained CNN and MobileNetV2 models should provide reliable facial drowsiness and eye-state predictions for the supported input conditions.

NFR3: Responsiveness

The system should transmit and display updated detection results through the web interface without significant communication delay.

NFR4: Usability

The web-based monitoring interface should clearly present the driver's current state, drowsiness score, eye status, and alert condition in an easily understandable manner.

NFR5: Maintainability

The system should maintain separate components for video processing, model inference, score fusion, backend communication, and frontend visualization to simplify future modifications.

NFR6: Non-Contact Operation

The system should perform driver monitoring using a webcam without requiring wearable devices or physical physiological sensors.

NFR7: Extensibility

The system architecture should support future integration of additional driver-monitoring features, deep learning models, and vehicle safety mechanisms.

4. Project Design and Implementation

4.1 Proposed Project Architecture

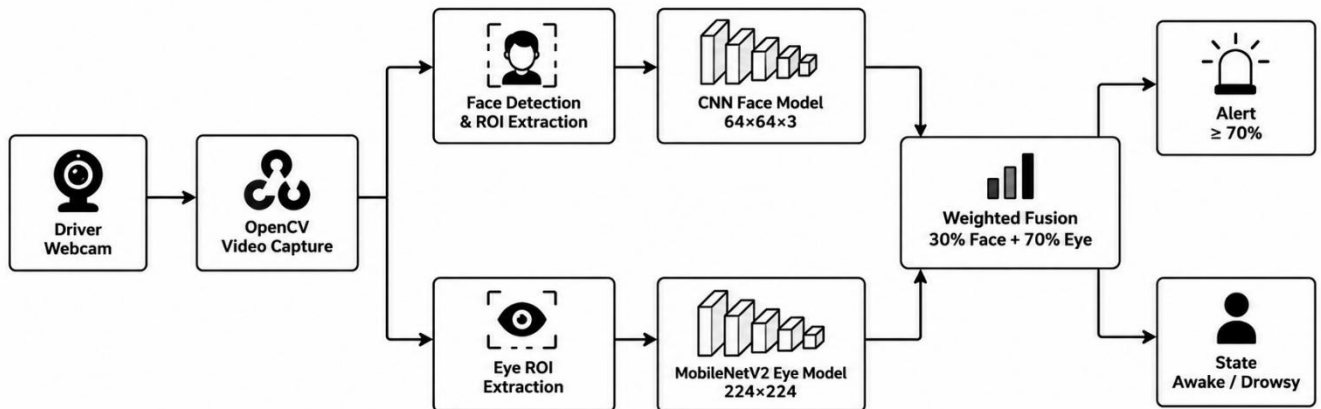


Figure 4.1: Proposed Project Architecture of Driver Drowsiness Detection Using CNN.

The proposed Driver Drowsiness Detection System follows an integrated real-time architecture that connects webcam-based image acquisition, OpenCV-based face processing, deep learning inference, weighted score fusion, backend communication, and web-based visualization. As shown in Figure 4.1, the driver's live video is first captured through the webcam and passed to the OpenCV video-processing module. OpenCV detects the driver's face and extracts the corresponding facial region. The detected face is then processed by the CNN-based facial drowsiness model, which produces a face-based drowsiness score. At the same time, the relevant eye regions are extracted from the detected face and passed to the MobileNetV2-based eye-state model, which determines the eye condition as open or closed and provides the eye-based drowsiness information. Thus, the architecture uses two complementary visual analysis paths: one for overall facial drowsiness classification and another focused specifically on eye state.

The outputs from the CNN and MobileNetV2 models are forwarded to the weighted score-fusion module, where the final drowsiness score is calculated using 30% contribution from the face-based score and 70% contribution from the eye-based score. The resulting score is compared with the predefined 70% threshold; a score equal to or above this threshold produces a drowsy condition and corresponding alert state, while a score below the threshold represents the non-drowsy state. The processed detection information is then handled by the FastAPI backend and transmitted to the web interface through WebSocket communication. The React-based dashboard receives these real-time results and presents the driver's alertness state, drowsiness score, eye condition, and alert information. Therefore, the architecture provides an end-to-end flow from live video capture → face and eye analysis → deep learning prediction → weighted score fusion → drowsiness decision → backend communication → real-time dashboard visualization, as represented in Figure 4.1.

4.2 High Level Design

The high-level design of the proposed Driver Drowsiness Detection System represents the major modules of the system and the flow of information between them. The architecture follows a real-time backend-driven processing approach, where the driver's webcam is accessed by the Python-based detection engine using OpenCV. The captured video frames are continuously processed to detect the driver's face and obtain the facial and eye regions required for deep learning-based analysis.

The detection engine consists of two parallel model-processing paths. The detected facial region is provided to the trained CNN-based facial drowsiness model, which produces a face-based drowsiness score. The relevant eye regions are processed by the MobileNetV2-based eye-state model, which classifies the eyes as open or closed and provides the corresponding eye-based drowsiness information. The outputs from these two models are combined by the weighted score-fusion module, using 30% weight for the face-based score and 70% weight for the eye-based score. The resulting final drowsiness score is then compared with the predefined 70% threshold to determine the driver's alertness state and whether an alert condition should be generated.

The processed results are passed to the FastAPI backend, which provides the communication layer between the detection engine and the web application. The detection results, including the final drowsiness score, driver state, eye status, and alert information, are transmitted to the frontend using WebSocket communication. The React-based monitoring dashboard receives these real-time updates and presents the information in a user-friendly interface. Therefore, the overall architecture follows the data flow:

Webcam → OpenCV Video Processing → CNN and MobileNetV2 Models → Weighted Score Fusion → Drowsiness Decision → FastAPI Backend → WebSocket → React Dashboard.

Major Components

1. Webcam and Video Acquisition:

The webcam provides the live video stream of the driver. The Python-based detection engine continuously captures video frames for real-time analysis.

2. OpenCV Video Processing:

OpenCV handles video-frame acquisition and face detection. The detected face is used to obtain the facial region and relevant eye regions for subsequent model processing.

3. CNN-Based Facial Drowsiness Model:

The extracted facial region is prepared according to the CNN input requirement of $64 \times 64 \times 3$ and processed to obtain a facial drowsiness score.

4. MobileNetV2-Based Eye-State Model:

The extracted eye regions are prepared according to the model input requirement of 224×224 and processed to classify the eyes as open or closed and obtain the eye-based drowsiness information.

5. Weighted Score-Fusion Module:

The face-based and eye-based outputs are combined into a single drowsiness score using the defined fusion weights:

$$\text{Final Drowsiness Score} = 0.3 \times \text{Face Score} + 0.7 \times \text{Eye Score}$$

6. Drowsiness Decision and Alert:

The final drowsiness score is compared with the 70% threshold. A score equal to or greater than 70% results in the drowsy/alert condition, while a score below the threshold corresponds to the non-drowsy state.

7. FastAPI Backend:

FastAPI provides the backend service responsible for handling the processed detection information and communication between the detection engine and frontend.

8. WebSocket Communication:

WebSocket provides continuous real-time transmission of processed detection results from the backend to the React frontend.

9. React-Based Monitoring Dashboard:

The dashboard presents the real-time detection information, including the driver state, drowsiness score, eye status, and alert information.



Figure 4.2: High Level Design and Data Flow.

4.3 Detailed Design

The detailed design describes the processing sequence used to detect driver drowsiness from live webcam video. The system combines OpenCV-based face and eye-region extraction with CNN and MobileNetV2 models, followed by weighted score fusion and threshold-based decision making.

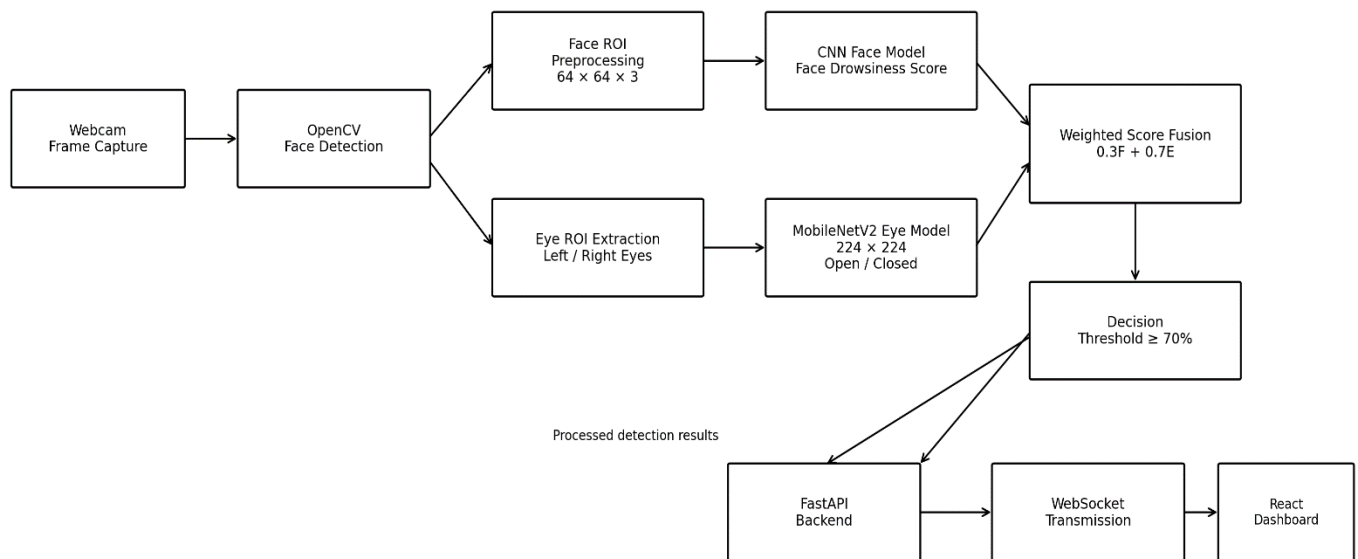


Figure 4.3: Detailed Design of Driver Drowsiness Detection Using CNN.

- Video Acquisition and Face Detection:

The webcam continuously captures the driver's video, while OpenCV processes the incoming frames. The driver's face is detected and the facial region is extracted. The detected face is also used to obtain the relevant eye regions.

- CNN-Based Facial Classification:

The extracted facial region is resized to $64 \times 64 \times 3$ and processed by the trained CNN model to obtain a face-based drowsiness score.

- MobileNetV2-Based Eye-State Classification:

The extracted eye regions are resized to 224×224 and processed using the MobileNetV2-based eye-state model. The model classifies the eyes as Open or Closed and provides the eye-based drowsiness information.

- Weighted Score Fusion:

The face-based and eye-based predictions are combined using the defined weighted fusion mechanism:

$$\text{Final Drowsiness Score} = 0.3 \times \text{Face Score} + 0.7 \times \text{Eye Score}$$

The face prediction contributes 30%, while the eye-based prediction contributes 70% to the final score.

- Drowsiness Decision:

The final drowsiness score is compared with the predefined 70% threshold:

- Score < 70%: Awake/Non-Drowsy
- Score $\geq$ 70%: Drowsy/Alert

- Backend and Real-Time Communication:

The processed detection results, including the drowsiness score, driver state, eye status, and alert condition, are handled by the FastAPI backend and transmitted to the React frontend through WebSocket communication.

- Dashboard Visualization:

The React-based dashboard displays the received results in real time, including the driver's current state, drowsiness score, eye status, and alert information.

4.4 Dataset Descriptions

The proposed system uses two image-based datasets for training its deep learning models. The Driver Drowsiness Dataset (DDD) is used for facial drowsiness classification, while the Open-Closed Eyes Dataset is used for eye-state classification. The two datasets support the separate face and eye analysis branches of the proposed detection pipeline.

4.4.1 Driver Drowsiness Dataset

The Driver Drowsiness Dataset (DDD) is used to train the CNN-based facial drowsiness classification model. It contains facial images representing different driver alertness conditions. The documented dataset distribution is shown in Table 4.1.

Table 4.1: Driver Drowsiness Dataset Distribution

Dataset Split	Number of Images
Training Set	33,435
Validation Set	8,358
Total	41,793

The facial images are processed according to the CNN model input requirement of $64 \times 64 \times 3$. The trained CNN model learns facial characteristics associated with drowsy and non-drowsy conditions and produces a face-based drowsiness score during inference.

4.4.2 Open-Closed Eyes Dataset

The Open-Closed Eyes Dataset is used to train the MobileNetV2-based eye-state classification model. The dataset contains images representing open and closed eye conditions. Its documented distribution is presented in Table 4.2.

Table 4.2: Open-Closed Eyes Dataset Distribution

Dataset Split	Number of Images
Training Set	139,804
Validation Set	27,961
Test Set	6,991
Total	174,756

The eye images are processed using an input resolution of 224×224 . The MobileNetV2-based model classifies the extracted eye regions as Open or Closed and provides the eye-based drowsiness information used by the fusion module.

4.4.3 Dataset Utilization in the Proposed System

The two datasets are used for different stages of the proposed driver drowsiness detection pipeline. The Driver Drowsiness Dataset (DDD) is used to train the CNN-based facial drowsiness classifier, which analyzes the overall facial region and produces a face-based drowsiness score. The Open-Closed Eyes Dataset is used to train the MobileNetV2-based eye-state classifier, which analyzes the extracted eye regions and classifies them as open or closed.

During real-time operation, the webcam captures the driver's face and the relevant facial and eye regions are extracted using OpenCV. The facial region is processed by the CNN model, while the eye regions are processed by the MobileNetV2 model. The outputs from both models are then combined using the defined weighted score-fusion mechanism, with 30% contribution from the face score and 70% from the eye score. The resulting score is used to determine the driver's final drowsiness state based on the 70% decision threshold.

Table 4.3: Dataset and Model Utilization

Dataset	Model	Input Size	Classification Task	Purpose
Driver Drowsiness Dataset	CNN	$64 \times 64 \times 3$	Drowsy / Non-Drowsy	Facial drowsiness classification
Open-Closed Eyes Dataset	MobileNetV2	224×224	Open / Closed	Eye-state classification

4.5 Tools and Technologies Used for Implementation

The proposed Driver Drowsiness Detection System integrates programming languages, deep learning frameworks, computer vision libraries, backend technologies, frontend technologies, and development tools.

4.5.1 Programming Languages

Python:

Python is the primary programming language for the detection pipeline. It is used for webcam processing, OpenCV-based face detection, CNN and MobileNetV2 model inference, score fusion, drowsiness decision logic, and backend integration.

JavaScript:

JavaScript is used to implement the interactive web-based frontend. It handles dashboard components and the real-time detection information received from the backend.

4.5.2 Deep Learning Frameworks

TensorFlow / Keras:

TensorFlow with Keras is used for the trained CNN-based facial drowsiness classification model. The model processes facial images with an input size of $64 \times 64 \times 3$ and produces a facial drowsiness prediction.

PyTorch:

PyTorch is used for the MobileNetV2-based eye-state model. The model processes eye images with an input size of 224×224 and classifies them as Open or Closed.

4.5.3 Computer Vision Library

OpenCV:

OpenCV forms the main computer vision layer of the system. It is used for webcam video acquisition, frame processing, face detection, and extraction of the facial and eye regions required for deep learning inference.

4.5.4 Backend and Communication Technologies

FastAPI:

FastAPI provides the backend service that connects the Python-based detection engine with the web-based monitoring interface. It handles the processed detection information generated by the detection pipeline.

WebSocket:

WebSocket provides continuous real-time communication between the FastAPI backend and React frontend. It transmits information such as the drowsiness score, driver state, eye status, and alert condition.

4.5.5 Frontend Technologies

React:

React is used to develop the web-based monitoring dashboard. It provides a component-based structure for displaying real-time driver monitoring information.

Vite:

Vite is used as the development and build tool for the React application, providing the environment required to develop and run the frontend.

Tailwind CSS:

Tailwind CSS is used to design and style the monitoring dashboard and provide a consistent responsive user interface.

Lucide Icons:

Lucide Icons are used to provide interface icons and visual indicators within the dashboard.

Recharts:

Recharts is used for graphical visualization of monitoring information, such as the drowsiness-score trend.

4.5.6 Development and Version Control Tools

Visual Studio Code:

Visual Studio Code is used as the primary development environment for writing, modifying, testing, and debugging the project source code.

Git:

Git is used for source-code version control and tracking changes made during project development.

GitHub:

GitHub is used for remote repository management, source-code storage, and maintaining different versions of the project.

5. Results and Discussion

5.1 Experimentation Environment Setup

The experimentation environment was configured with the required hardware and software components for developing and testing the proposed Driver Drowsiness Detection system. Python was used as the primary programming language, with TensorFlow/Keras for the CNN-based facial classification model and PyTorch with MobileNetV2 for eye-state classification. OpenCV was used for real-time video processing and face detection. The integrated system uses FastAPI as the backend, WebSocket for real-time communication, and React for the monitoring dashboard. Jupyter Notebook was used for model training and experimentation, while supporting libraries such as NumPy, Matplotlib, and Scikit-learn were used for data processing, visualization, and performance evaluation.

Table 5.1: Hardware Requirements

Component	Requirement
Processor	Intel Core i5 or equivalent
RAM	8 GB or above
Storage	512 GB or Sufficient storage for datasets, models and dependencies
GPU	Optional; NVIDIA GPU recommended for faster model training
Camera	Webcam or compatible camera for real-time detection
Internet	Required for dataset and package installation

Table 5.2: Software Requirements

Software	Purpose
Python 3.11 / 3.12	Programming and model development
TensorFlow / Keras	CNN model development and inference
PyTorch	Eye-state model implementation
MobileNetV2	Eye-state classification
OpenCV	Face/eye detection and real-time video processing
Scikit-learn	Evaluation metrics and machine-learning models
NumPy	Numerical computation and array processing
Matplotlib	Training and performance visualization
Jupyter Notebook	Model training and experimentation
FastAPI	Backend and real-time server
WebSocket	Real-time frontend-backend communication

React	Web-based dashboard development
Git / GitHub	Version control and project management

5.2 Functional Requirements Results Achieved

The implemented system was tested for its major functional requirements, including video acquisition, face detection, facial drowsiness classification, eye-state classification, score fusion, alert generation, and real-time dashboard visualization. The system combines the facial CNN score and eye-state score using a 30% face and 70% eye weighting, with a 70% combined-score threshold for drowsiness alert generation.

Model Evaluation Results:

Table 5.3: Evaluation Results

Model	Evaluation Set	Accuracy	Precision	Recall	F1-Score	AUC
CNN	Validation (8,358)	99.99%	100%	99.97%	99.99%	1.0000
MobileNetV2	Test (6,991)	99.41%	99.46%	99.77%	99.62%	0.9998

The CNN obtained 99.99% validation accuracy, while the MobileNetV2 eye-state classifier obtained 99.41% accuracy on the held-out test set. These results indicate high classification performance on their respective evaluation datasets.

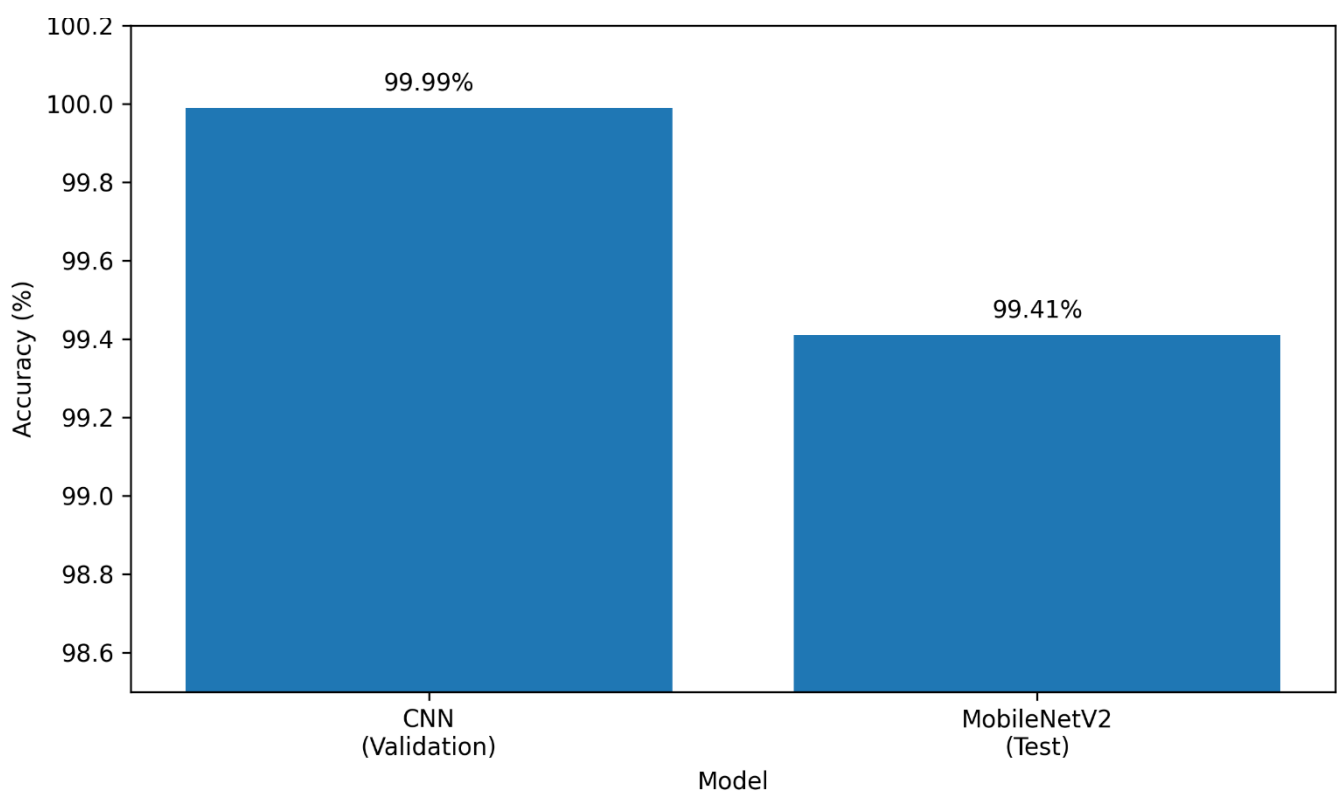


Figure 5.1: Accuracy of the Implemented Deep Learning Models.

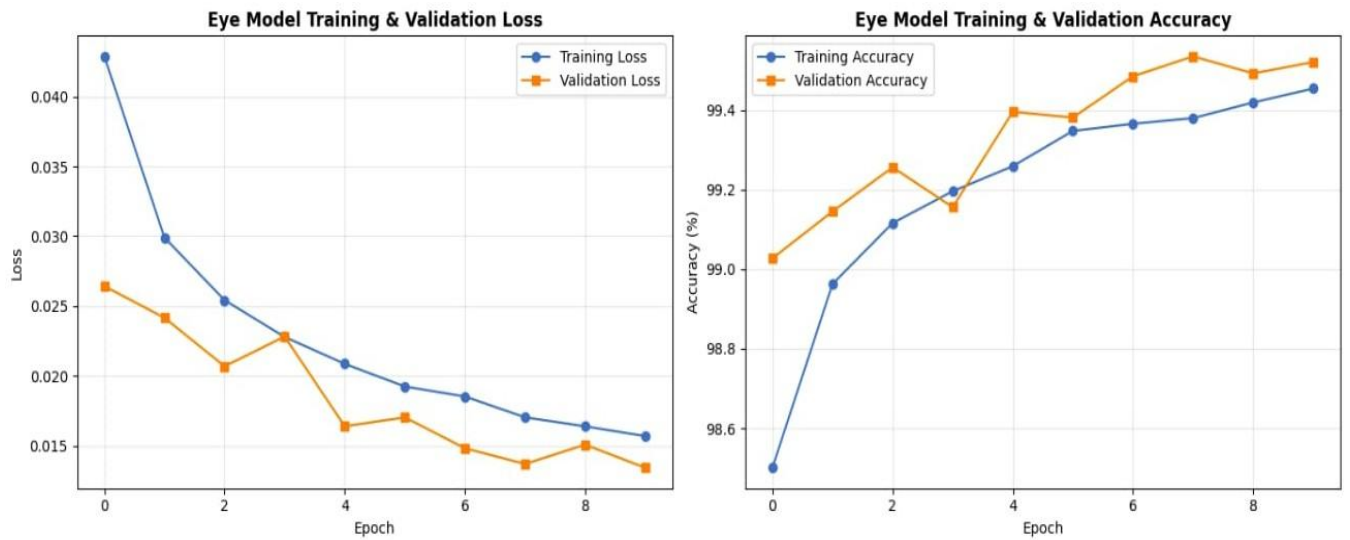


Figure 5.2: Eye Model Training and Validation Performance

Figure 5.2 shows the training and validation loss and accuracy of the MobileNetV2 eye-state model across the training epochs. The decreasing loss and increasing accuracy indicate effective learning of the eye-state classification task.

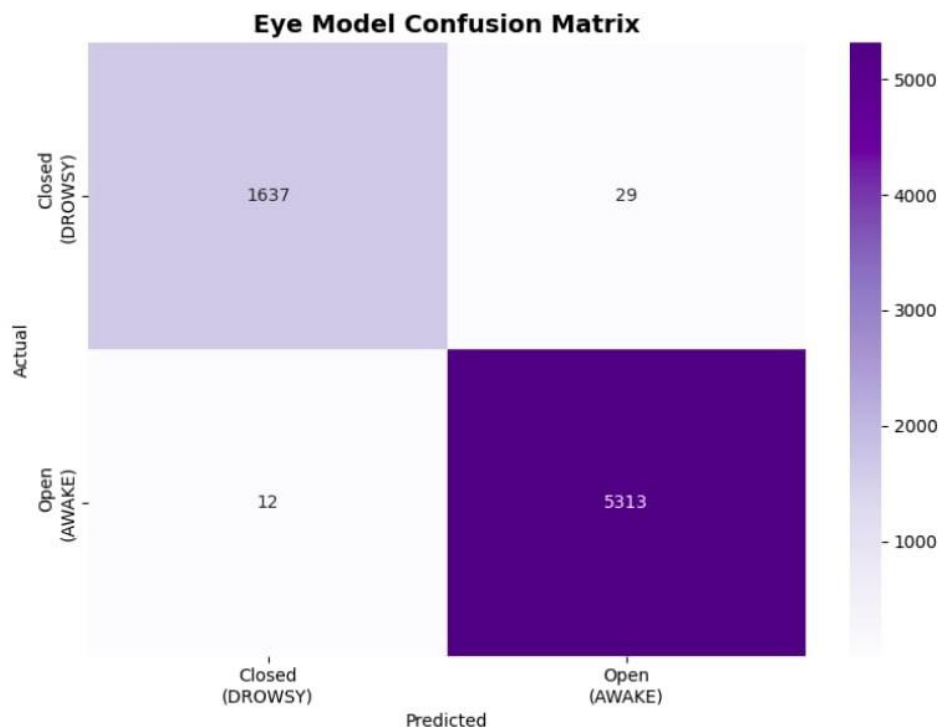


Figure 5.3: Confusion Matrix of the Eye-State Classification Model

Figure 5.3 presents the confusion matrix obtained on the eye-model test set. The model correctly classified 1,637 closed-eye samples and 5,313 open-eye samples, with 29 and 12 misclassified samples respectively.

Functional Test Results:

Table 5.4: Functional Requirements Results

Functional Requirement	Expected Result	Result
Webcam acquisition	Live video frames are captured	Frames captured successfully
Face detection	Driver's face is identified from the video frame	Face detected successfully
CNN prediction	Facial drowsiness score is generated	Score generated
Eye-state detection	Left and right eye states are identified	Eye states detected
Score fusion	Face and eye predictions are combined	Weighted score calculated
Drowsiness decision	Combined score is compared with the defined threshold	Drowsiness status determined
Alert generation	Alert is triggered when the threshold condition is met	Alert generated when required
Dashboard	Detection information is displayed to the user	Status displayed in real time
Real-time communication	Detection results are transferred to the frontend	Data transmitted through WebSocket

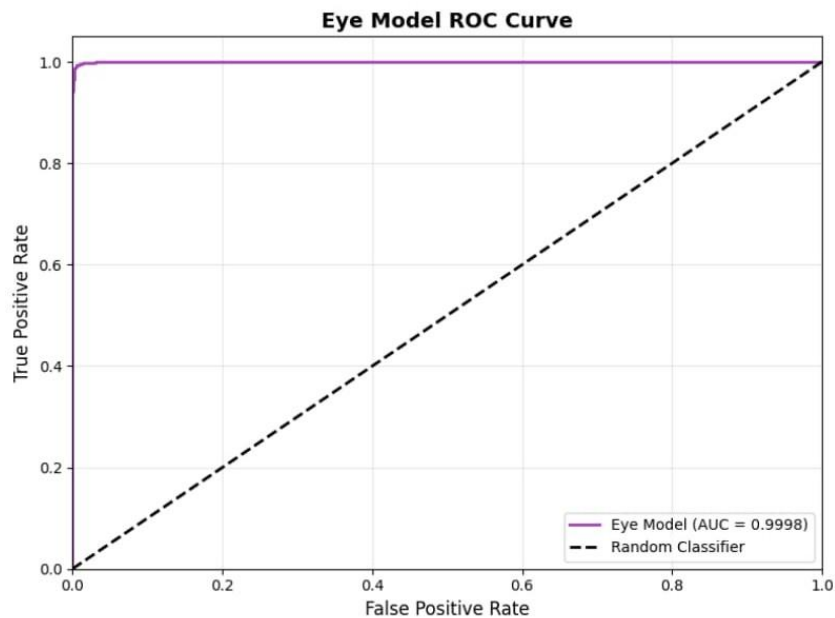


Figure 5.4: ROC Curve of the Eye-State Classification Model

Figure 5.4 shows the ROC curve of the MobileNetV2 eye-state classifier. The model obtained an AUC of 0.9998, indicating strong discrimination between the closed-eye and open-eye classes on the test set.

5.3 Non-Functional Requirements Results Achieved

The system was evaluated qualitatively for real-time operation, usability and reliability. A formal latency/FPS benchmark was not established in the current experiments, so numerical performance values should not be claimed.

Table 5.5: Non Functional Requirements Results

Non Functional Requirement	Observation	Result
Real-time operation	Continuous webcam-based detection	Supported
Responsiveness	Prediction results are continuously updated	Observed during operation
Usability	Dashboard provides visual monitoring and alerts	Implemented
Reliability	Models load and operate during live detection	Verified during testing
Maintainability	ML, backend, and frontend components are separated	Supported by system structure
Portability	Python-based implementation with documented dependencies	Supported
Performance benchmark	Formal latency and FPS measurements	Not measured

The non-functional evaluation focused on real-time operation, responsiveness, usability, reliability, maintainability, and portability of the implemented system. The system supports continuous webcam-based monitoring and provides real-time prediction updates through the dashboard. The separation of the machine-learning, backend, and frontend components supports maintainability. However, formal performance measurements such as average FPS, inference latency, and end-to-end response time were not conducted, so numerical performance claims are not reported.

5.4 Comparison of Results with Existing Works

Several existing studies have applied CNN and transfer-learning approaches for driver drowsiness detection. Reported performance varies with the dataset, classification task, and evaluation protocol. For example, Majeed et al. reported an average accuracy of 96.69% using a CNN-based approach on the YawDD dataset.

Civik and Yuzgec developed a real-time CNN-based system on the YawDD dataset and reported 93.6% accuracy for eye classification and 94.5% for mouth classification, with an average operating speed of 6 FPS on a Jetson Nano.

Salem and Waleed evaluated CNN and transfer-learning models across several datasets. On the Driver-Drowsiness dataset, their CNN achieved 99.69%, while MobileNetV2 achieved 99.94% accuracy.

Table 5.6: Comparison with Existing Works

Study	Method	Dataset	Reported Accuracy	Key Feature
Majeed et al. (2023)	CNN	YawDD	96.69%	Yawning/behavior-based drowsiness detection
Civik & Yuzgec (2023)	CNN	YawDD	93.6% Eye / 94.5% Mouth	Real-time embedded implementation
Salem & Waleed (2024)	CNN	Driver-Drowsiness Dataset	99.69%	CNN-based facial drowsiness classification
Salem & Waleed (2024)	MobileNetV2	Driver-Drowsiness Dataset	99.94%	Transfer-learning-based classification
Proposed System - CNN	CNN	DDD	99.99%*	Facial drowsiness classification
Proposed System MobileNetV2	MobileNetV2	Open-Closed Eyes	99.41%†	Eye-state classification
Proposed System	CNN + MobileNetV2 Fusion	DDD + Eye Dataset	-	30:70 weighted score fusion and real-time dashboard

The proposed system uses two complementary models: a CNN for facial drowsiness classification and MobileNetV2 for eye-state classification. The two predictions are combined using a 30% face and 70% eye weighting, followed by a drowsiness threshold. This provides a multimodel approach rather than relying on a single visual indicator.

The obtained CNN validation accuracy of 99.99% and MobileNetV2 test accuracy of 99.41% are high, but these values should not be interpreted as a direct numerical ranking over previous studies, because the datasets, splits, classification tasks, and evaluation procedures are different. This is consistent with recent

literature noting that direct cross-study accuracy comparisons require caution when evaluation conditions differ.

5.5 Project GUI Snapshots

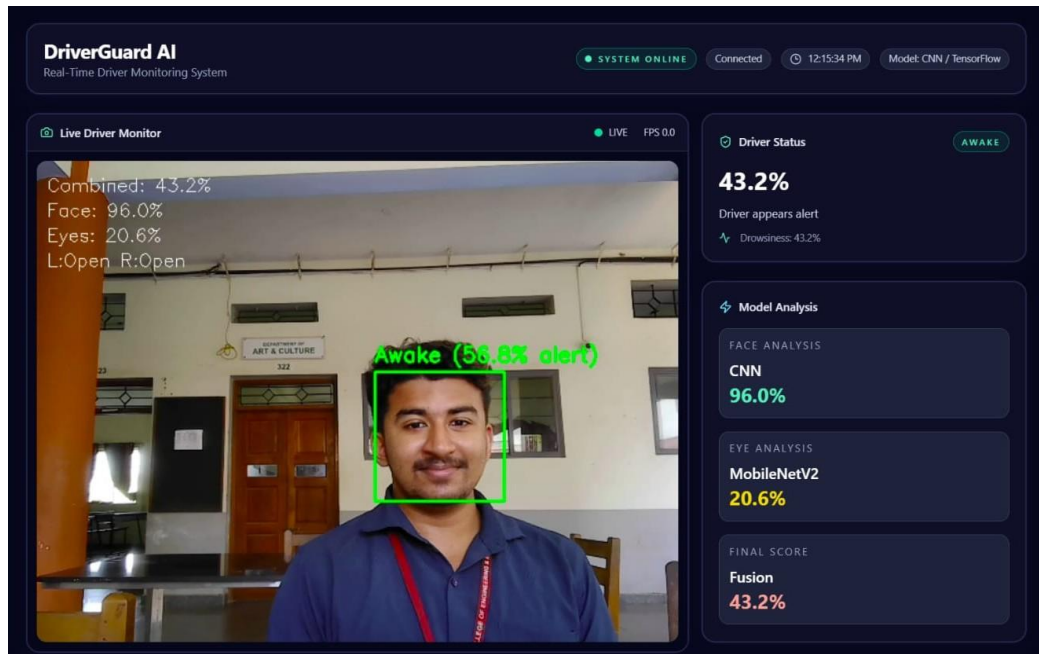


Figure 5.5: Main Dashboard and Awake State Monitoring

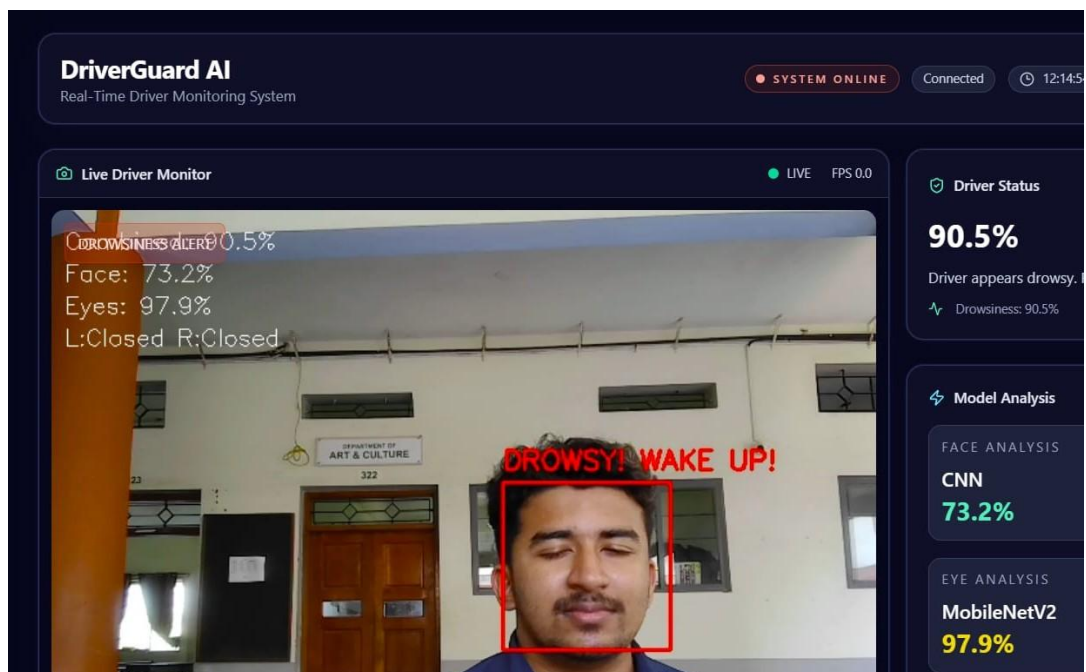


Figure 5.6: Drowsiness Detection and Alert Generation

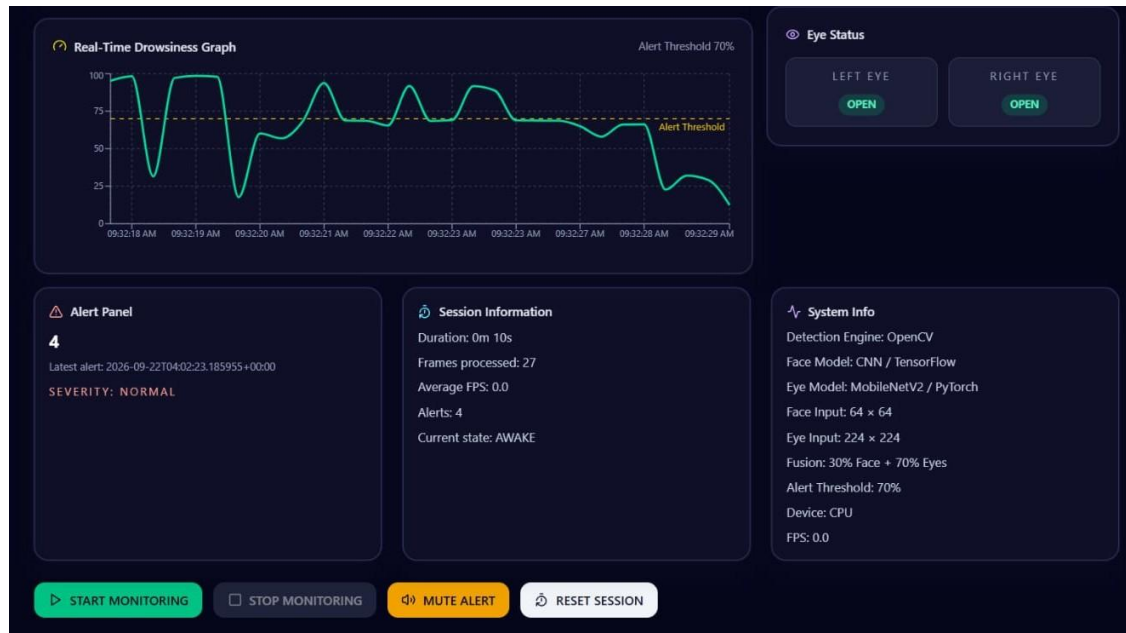


Figure 5.7: Real-Time Drowsiness Analysis and Session Monitoring

5.6 Societal Impact of the Project

Driver drowsiness can reduce driver alertness and reaction capability. The proposed system provides continuous camera-based monitoring and generates an alert when the estimated drowsiness level crosses the defined threshold.

Potential applications include:

- Long-distance driving
- Commercial and fleet transportation
- Night-time driving
- Driver-assistance systems
- Safety monitoring

The system is intended as a driver-assistance and warning mechanism, rather than a replacement for human judgment. Camera positioning, lighting, facial occlusion and individual differences may affect detection performance.

5.7 SDG Mapped

SDG 3: Good Health and Well-Being

The project supports road-safety-related well-being by providing a mechanism for detecting possible driver drowsiness and issuing timely alerts.

SDG 8: Decent Work and Economic Growth

For commercial transportation, driver-monitoring systems can potentially support safer working conditions for professional drivers.

SDG 9: Industry, Innovation and Infrastructure

The project applies deep learning, computer vision and real-time AI to an intelligent transportation/safety application.

SDG 11: Sustainable Cities and Communities

The system can potentially contribute to safer transportation environments through driver-monitoring technology.

Conclusion and Future Scope

The Driver Drowsiness Detection using CNN project successfully demonstrates a real-time approach for identifying driver drowsiness using facial and eye-state analysis. The system integrates a CNN-based facial drowsiness classifier with a MobileNetV2-based eye-state classifier to improve monitoring through complementary visual information. The face and eye predictions are combined using a 30% face and 70% eye weighting, with a 70% threshold for generating drowsiness alerts. The CNN achieved 99.99% validation accuracy, while the MobileNetV2 model achieved 99.41% accuracy on the held-out test set. The developed React dashboard with FastAPI and WebSocket communication provides real-time visualization of the camera feed, model predictions, eye status, drowsiness score, alerts, and session information. Thus, the project provides an integrated software solution for driver monitoring and safety assistance.

Future Scope

The system can be further improved by evaluating it under diverse real-world conditions such as different lighting environments, camera positions, facial appearances, and partial face occlusions. A larger and more diverse dataset can be used to improve generalization. Future versions can incorporate additional behavioral indicators such as head pose, yawning, gaze direction, and prolonged eye closure. Performance can also be optimized for edge devices and embedded platforms to support practical vehicle deployment. Formal measurements of FPS, inference latency, memory usage, and CPU/GPU utilization can be conducted to evaluate deployment efficiency. Future work may also include driver-specific calibration, improved alert mechanisms, cloud-based monitoring, and integration with vehicle safety systems.

References

- [1] S. A. El-Nabi, A. F. Ibrahim, H. H. Abo Emira, S. Alshathri, E.-S. M. El-Rabaie, A. Emam, and W. El-Shafai, "Dynamic driver drowsiness detection with attention enhanced convolutional neural networks for real time monitoring and road safety applications," *Scientific Reports*, vol. 16, Art. no. 26340, June 2026.
- [2] F. Wei, J. Yang, Y. Wang, L. Lin, and H. Zhang, "Prior knowledge-guided multi-information graph convolutional network for driver drowsiness detection," *Expert Systems with Applications*, vol. 275, Art. no. 127028, May 2025.
- [3] S. Abd El-Nabi, K. F. Ramadan, E.-S. M. El-Rabaie, A. Emam, and W. El-Shafai, "A real-time design and implementation of intelligent drowsiness and fatigue recognition system for enhancing driver safety," *Engineering Applications of Artificial Intelligence*, vol. 162, Art. no. 112665, December 2025.
- [4] H. Beles, T. Vesselenyi, A. Rus, T. Mitran, F. B. Scurt, and B. A. Tolea, "Driver drowsiness multi-method detection for vehicles with autonomous driving functions," *Sensors*, vol. 24, no. 5, Art. no. 1541, February 2024.
- [5] L. Lin, S. Wang, J. Yang, and F. Wei, "A multi-aware graph convolutional network for driver drowsiness detection," *Knowledge-Based Systems*, vol. 305, Art. no. 112643, December 2024.
- [6] D. Salem and M. Waleed, "Drowsiness detection in real-time via convolutional neural networks and transfer learning," *Journal of Engineering and Applied Science*, vol. 71, Art. no. 122, May 2024.
- [7] S. Pahariya, P. Vats, and S. Suchitra, "Driver drowsiness detection using MobileNetV2 with transfer learning approach," in *Proc. 2024 International Conference on Advances in Data Engineering and Intelligent Computing Systems (ADICS)*, August 2024.
- [8] R. Florez, F. Palomino-Quispe, R. J. Coaquira-Castillo, J. C. Herrera-Levano, T. Paixão, and A. B. Alvarez, "A CNN-based approach for driver drowsiness detection by real-time eye state identification," *Applied Sciences*, vol. 13, no. 13, Art. no. 7849, July 2023.
- [9] I. Jahan, K. M. A. Uddin, S. A. Murad, S. M. U. Miah, T. Z. Khan, M. Masud, S. Aljahdali, and A. K. Bairagi, "4D: A real-time driver drowsiness detector using deep learning," *Electronics*, vol. 12, no. 1, Art. no. 235, January 2023.
- [10] I. D. B. Rozaimi, F. H. K. Zaman, S. A. C. Abdullah, H. Z. Abidin, et al., "Driver drowsiness detection using deep learning models based on different camera positions," in *Proc. 11th International Conference on Information and Communication Technology (ICoICT)*, August 2023.
- [11] V. Kumar, S. Sharma, and R. Walia, "Driver drowsiness detection using modified deep learning architecture," *Evolutionary Intelligence*, vol. 16, no. 6, pp. 1907–1916, June 2022.
- [12] M. Ahmed, S. Masood, M. Ahmad, and A. A. Abd El-Latif, "Intelligent driver drowsiness detection for traffic safety based on multi CNN deep model and facial subsampling," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 10, pp. 19743–19752, October 2022.
- [13] S. E. Bekhouche, Y. Ruichek, and F. Dornaika, "Driver drowsiness detection in video sequences using hybrid selection of deep features," *Knowledge-Based Systems*, vol. 252, Art. no. 109436, September 2022.
- [14] E. Magán, M. P. Sesmero, J. M. Alonso-Weber, and A. Sanchis, "Driver drowsiness detection by applying deep learning techniques to sequences of images," *Applied Sciences*, vol. 12, no. 3, Art. no. 1145, January 2022.

- [15] M. Vimala, M. Nandhini, S. Jasmine, P. Ranjith Kumar, and S. Ramasamy, "Implementation of a lightweight deep learning model for detecting driver fatigue," *Franklin Open*, vol. 15, Art. no. 100547, June 2026.
- [16] M. W. Gomaa, R. O. Mahmoud, and A. M. Sarhan, "A CNN-LSTM-based deep learning approach for driver drowsiness prediction," *Journal of Engineering Research*, vol. 6, no. 3, pp. 59–70, 2022.
- [17] M. Irsan, R. Hassan, A. H. Ibrahim, M. K. Hasan, M. C. Lam, and W. M. H. W. Hussain, "A framework for driver drowsiness monitoring using a convolutional neural network and the Internet of Things," *Intelligent Automation & Soft Computing*, vol. 39, no. 2, pp. 157–174, May 2024.
- [18] A.-C. Phan, T.-N. Trieu, and T.-C. Phan, "Driver drowsiness detection and smart alerting using deep learning and IoT," *Internet of Things*, vol. 22, Art. no. 100705, July 2023.
- [19] T. Fonseca and S. Ferreira, "Drowsiness detection in drivers: A systematic review of deep learning-based models," *Applied Sciences*, vol. 15, no. 16, Art. no. 9018, August 2025.
- [20] M. Venkateswarlu and V. R. R. Chirra, "CNN-ViT: A multi-feature learning based approach for driver drowsiness detection," *Array*, vol. 27, Art. no. 100425, September 2025.